

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

September 21, 2026

Outline

- 1 SoPS Lecture 8
 - Break the Cube
 - Wrapping up

Outline

- 1 SoPS Lecture 8
 - Break the Cube
 - Wrapping up

Review

Last time

Review

Last time

- we explored Battleship firing algorithms

Review

Last time

- we explored Battleship firing algorithms
- we talked about static vs. dynamic algorithms

Review

Last time

- we explored Battleship firing algorithms
- we talked about static vs. dynamic algorithms
- we found good patterns for static algorithms

Break the Cube

Activity (60 mins)

Break the Cube

Activity (60 mins)

- split into groups of up to 4

Break the Cube

Activity (60 mins)

- split into groups of up to 4
- take some time to read the rules

Break the Cube

Activity (60 mins)

- split into groups of up to 4
- take some time to read the rules
- play a few games

Break the Cube

Activity (60 mins)

- split into groups of up to 4
- take some time to read the rules
- play a few games
- **record data:** how many clues until a correct guess

Break the Cube: Debrief

Break the Cube: Debrief

Let's combine the data ...

Break the Cube: Debrief

Let's combine the data ...

Question

How many clues do we need to find a solution?

Question

What kinds of boards took the most clues?

Break the Cube: Solitaire

Problem

Suppose we have the clues:

- B: green, blank, blank
- C: yellow, gray, blank
- J: green, blank, blank

with gray=small, green=medium, and yellow=large.

Find the block setup.

Prove it!

Prove it!

Proof

A proof is a logical argument that explains why a statement must be true no matter what.

Prove it!

Proof

A proof is a logical argument that explains why a statement must be true no matter what.

Example:

For any integer n , the number $n^2 + n$ is even.

Prove it!

Proof

A proof is a logical argument that explains why a statement must be true no matter what.

Example:

For any integer n , the number $n^2 + n$ is even.

Proof:

If n is an integer, either n or $n + 1$ will be even.

Therefore $n^2 + n = n(n + 1)$ is the product of an even number with another number.

Any number times an even number is even.

Therefore $n^2 + n$ must be even.

Build your own

Activity

Create a Break the Cube solitaire problem, meaning:

- specify the colors of the small, medium, large blocks
- enough clues to determine a **unique** setup

Build your own

Activity

Create a Break the Cube solitaire problem, meaning:

- specify the colors of the small, medium, large blocks
- enough clues to determine a **unique** setup

Activity

Swap your puzzle with another group.

- find a solution of their puzzle
- explain why the solution is unique (or find another)
- try to write a proof

Outline

- 1 SoPS Lecture 8
 - Break the Cube
 - Wrapping up

Discussion

Discussion

Question

What makes a good choice of setup for your Break the Cube board?

Discussion

Question

What makes a good choice of setup for your Break the Cube board?

Question

What does it take to *prove* that given a set of clues your solution is right?

Question of the Day

Question of the Day

Puzzle

Suppose we wanted to try to count the number of possible arrangements of the three blocks. How might we begin to go about it?

Final thoughts

Reflection

Final thoughts

Reflection

- Describe what a proof is in your own words.

Final thoughts

Reflection

- Describe what a proof is in your own words.
- Find an example of a proof online and write about it. What is it proving? What is the crux of the argument, in your own words?

Submit your responses by **Tonight!**